

Assignment

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Examples of Sets

Even numbers:

$$A = \{x \in \mathbb{Z} \mid x \text{ is even}\}$$

Odd numbers:

$$B = \{x \in \mathbb{Z} \mid x \text{ is odd}\}$$

Examples of Sets (contd.)

Natural numbers less than 10:

$$C = \{x \in \mathbb{N} \mid x < 10\}$$

Multiples of 5:

$$D = \{x \in \mathbb{Z} \mid x = 5n, n \in \mathbb{Z}\}$$

Examples of Sets (contd.)

Prime numbers:

$$E = \{x \in \mathbb{N} \mid x \text{ is prime}\}$$

Integers between -5 and 5:

$$F = \{x \in \mathbb{Z} \mid -5 \leq x \leq 5\}$$

Examples of Sets (contd.)

Perfect squares:

$$G = \{x \in \mathbb{N} \mid x = n^2, n \in \mathbb{N}\}$$

Real numbers greater than 0:

$$H = \{x \in \mathbb{R} \mid x > 0\}$$

Examples of Sets (contd.)

Students scoring above 80:

$$I = \{x \in S \mid \text{score}(x) > 80\}$$

Letters in the word APPLE:

$$J = \{x \in \{A, B, C, \dots, Z\} \mid x \text{ appears in "APPLE"}\}$$

Inclusion-Exclusion Principle

General Formula:

$$\left| \bigcup_{i=1}^{\infty} A_i \right| = \sum_{k=1}^{\infty} (-1)^{k+1} \left(\sum_{1 \leq i_1 < i_2 < \dots < i_k} |A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}| \right)$$

Proof Sketch

1. Finite Truncation:

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} |A_{i_1} \cap \dots \cap A_{i_k}|$$

2. Limit Process: As $n \rightarrow \infty$, union approaches infinite union.

Proof Sketch (contd.)

3. Element-wise Counting:

$$\binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \cdots + (-1)^{m+1} \binom{m}{m} = 1$$

4. Conclusion: Each element counted exactly once.

Types of Sets

Type of Set	Example
Numerical	Sensor readings: $\{23.1, 24.5, 25.0\}$
Categorical	Sentiment labels: $\{\text{Positive}, \text{Negative}\}$
Time Series	Stock prices: $\{(t_1, 100), (t_2, 102)\}$
Ordinal	Satisfaction: $\{\text{Poor}, \text{Fair}, \text{Good}\}$

Types of Sets (contd.)

Type of Set	Example
Image Set	MNIST digits $\{0, 1, 2, \dots, 9\}$
Text Set	NLP corpus: $\{\text{"AI is powerful"}\}$
Web/API Set	Tweets: $\{\text{Tweet}_1, \text{Tweet}_2\}$
Multivariate	Patient records: $\{(\text{Age}, \text{BP})\}$

Partially Ordered Sets (Posets)

Definition: A poset $(P, \leq)$ satisfies:

- ▶ Reflexive: $a \leq a$
- ▶ Antisymmetric: $a \leq b$ and $b \leq a \Rightarrow a = b$
- ▶ Transitive: $a \leq b, b \leq c \Rightarrow a \leq c$

Not all elements need to be comparable.

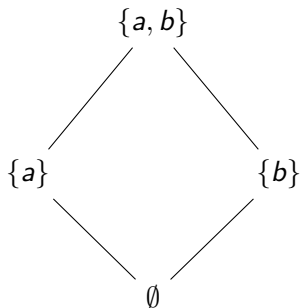
Poset Example: Divisibility on $\{1, 2, 4\}$

$$1 \leq 2 \leq 4$$



Poset Example: Power set of $\{a, b\}$

$$\emptyset \subseteq \{a\}, \{b\} \subseteq \{a, b\}$$



Poset Example: Divisibility on $\{1, 2, 3, 6\}$

$$1 \leq 2 \leq 6, \quad 1 \leq 3 \leq 6$$

